Hands-on Task: Run and Manage a "Hello Web App" (httpd)

Objective

Deploy and manage a simple Apache-based web server and:

- **Verify** it is running
- **Modify** it
- **Scale** it
- **Debug** it

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1. Deploying a Simple Pod

- **Run Pod:** `kubectl run apache-pod --image=httpd`

```
anirudh_chawla@LUCIF3R:~$ kubectl run apache-pod --image=httpd
pod/apache-pod created
anirudh_chawla@LUCIF3R:~$ kubectl get pods
NAME          READY   STATUS              RESTARTS   AGE
apache-pod    0/1     ContainerCreating   0          13s
```

- **Check Status:** `kubectl get pods`
- **Inspect:** `kubectl describe pod apache-pod`
  - Focus on the image (`httpd`), port (`80`), and events.

```
anirudh_chawla@LUCIF3R:~$ kubectl describe pod apache-pod
Name:             apache-pod
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Wed, 25 Mar 2026 06:02:09 +0000
Labels:           run=apache-pod
Annotations:      <none>
Status:           Running
IP:               10.244.0.3
IPs:
  IP:  10.244.0.3
Containers:
  apache-pod:
    Container ID:   docker://6fe8e53dde30a711bea68eeccfde7bd299eef6883c175ae57f682334429d7086
    Image:          httpd
    Image ID:       docker-pullable://httpd@sha256:331548c5249bdeced0f048bc2fb8c6b6427d2ec6508bed9c1fec6c57d0b27a60
    Port:           <none>
    Host Port:      <none>
    State:          Running
      Started:      Wed, 25 Mar 2026 06:02:28 +0000
    Ready:          True
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-nwmlm (ro)
```

- **Access:** `kubectl port-forward pod/apache-pod 8081:80`

```
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-nwmlm:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    ConfigMapOptional:       <nil>
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type    Reason     Age   From               Message
  ----    ------     ----  ----               -------
  Normal  Scheduled  72s   default-scheduler  Successfully assigned default/apache-pod to minikube
  Normal  Pulling    71s   kubelet            Pulling image "httpd"
  Normal  Pulled     53s   kubelet            Successfully pulled image "httpd" in 16.178s (16.178s including waiting).
Image size: 116780838 bytes.
  Normal  Created    53s   kubelet            Container created
  Normal  Started    53s   kubelet            Container started
```

  - Open `http://localhost:8081` to see "It works!".

It works!

- **Cleanup:** `kubectl delete pod apache-pod`

```
^Canirudh_chawla@LUCIF3R:~kubectl delete pod apache-pod
pod "apache-pod" deleted
```

  - Note: Pods do not self-heal; they disappear permanently when deleted.

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2. Transitioning to a Deployment

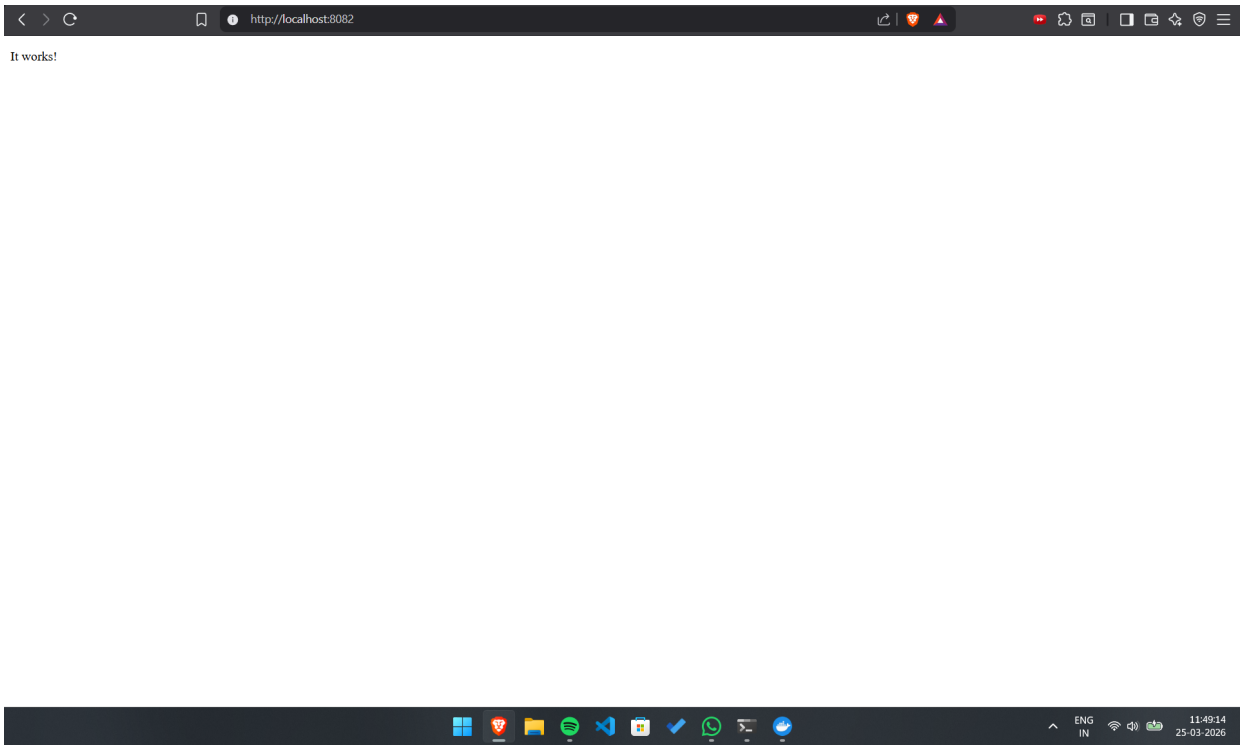
- **Create Deployment:** `kubectl create deployment apache --image=httpd`

```
anirudh_chawla@LUCIF3R:~$ kubectl create deployment apache --image=httpd
deployment.apps/apache created
anirudh_chawla@LUCIF3R:~$ kubectl get deployments
kubectl get pods
NAME      READY   UP-TO-DATE   AVAILABLE   AGE
apache    1/1     1            1           5s
NAME                      READY   STATUS    RESTARTS   AGE
apache-5959974c77-295l5   1/1     Running   0          5s
```

- **Expose App:** `kubectl expose deployment apache --port=80 --type=NodePort`

```
anirudh_chawla@LUCIF3R:~$ kubectl expose deployment apache --port=80 --type=NodePort
service/apache exposed
anirudh_chawla@LUCIF3R:~$ kubectl port-forward service/apache 8082:80
Forwarding from 127.0.0.1:8082 -> 80
Forwarding from [::1]:8082 -> 80
Handling connection for 8082
```

- **Access via Service:** `kubectl port-forward service/apache 8082:80`



3. Scaling

- **Scale Up:** `kubectl scale deployment apache --replicas=2`
  - This ensures multiple pods run the same application.

```
^Canirudh_chawla@LUCIF3R:~kubectl scale deployment apache --replicas=2
deployment.apps/apache scaled
anirudh_chawla@LUCIF3R:~$ kubectl get pods
NAME                      READY   STATUS    RESTARTS   AGE
apache-5959974c77-295l5   1/1     Running   0          2m14s
apache-5959974c77-vdj2l   1/1     Running   0          5s
```

```
anirudh_chawla@LUCIF3R:~$ kubectl set image deployment/apache httpd=wrongimage
deployment.apps/apache image updated
anirudh_chawla@LUCIF3R:~$ kubectl get pods
NAME                      READY   STATUS         RESTARTS   AGE
apache-5959974c77-295l5   1/1     Running        0          3m18s
apache-5959974c77-vdj2l   1/1     Running        0          69s
apache-7d5c895d55-lb6c7   0/1     ErrImagePull   0          6s
```

4. Debugging and Self-Healing

- **Simulate Failure:** Set the image to a non-existent one:

- `kubectl set image deployment/apache httpd=wrongimage`

```
anirudh_chawla@LUCIF3R:~$ kubectl describe pod apache-7d5c895d55-lb6c7
Name:             apache-7d5c895d55-lb6c7
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Wed, 25 Mar 2026 06:21:23 +0000
Labels:           app=apache
                  pod-template-hash=7d5c895d55
Annotations:      <none>
Status:           Pending
IP:               10.244.0.6
IPs:
  IP:           10.244.0.6
Controlled By:  ReplicaSet/apache-7d5c895d55
Containers:
  httpd:
    Container ID:
    Image:          wrongimage
    Image ID:
    Port:           <none>
    Host Port:      <none>
    State:          Waiting
      Reason:       ErrImagePull
    Ready:          False
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-dtt7r (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       False
  ContainersReady             False
  PodScheduled                True
Volumes:
  kube-api-access-dtt7r:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    ConfigMapOptional:       <nil>
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
```

- **Diagnose:** Look for `ImagePullBackOff` in `kubectl describe pod`.

```
anirudh_chawla@LUCIF3R:~$ kubectl set image deployment/apache httpd=httpd
deployment.apps/apache image updated
anirudh_chawla@LUCIF3R:~$ kubectl get pods
NAME                      READY   STATUS    RESTARTS   AGE
apache-5959974c77-295l5   1/1     Running   0          28m
apache-5959974c77-vdj2l   1/1     Running   0          26m
anirudh_chawla@LUCIF3R:~$ kubectl exec -it apache-5959974c77-vdj2l -- /bin/bash
root@apache-5959974c77-vdj2l:/usr/local/apache2# ls /usr/local/apache2/htdocs
index.html
root@apache-5959974c77-vdj2l:/usr/local/apache2# exit
exit
```

- **Fix:** Revert to the correct image: `kubectl set image deployment/apache httpd=httpd`.
- **Observe Healing:** Delete one pod and watch Deployment recreate it automatically.

```
anirudh_chawla@LUCIF3R:~$ kubectl delete pod apache-5959974c77-295l5
pod "apache-5959974c77-295l5" deleted
anirudh_chawla@LUCIF3R:~$ kubectl get pods -w
NAME                      READY   STATUS    RESTARTS   AGE
apache-5959974c77-gjwc7   1/1     Running   0          6s
apache-5959974c77-vdj2l   1/1     Running   0          29m
```

---

5. Terminal Management (Port-Forwarding)

- **Why it blocks:** It creates a live foreground tunnel.
- **Backgrounding:** Use `&` to run in background.

- **Management:**
  - Identify: `jobs` or `ps aux | grep port-forward`
  - Stop: `kill %1` or `kill <PID>`
  - Recommended: Use `tmux` for persistent sessions.

Final Cleanup

- `kubectl delete deployment apache`
- `kubectl delete service apache`

```
^Canirudh_chawla@LUCIF3R:~kubectl delete deployment apache
kubectl delete service apache
deployment.apps "apache" deleted
service "apache" deleted
anirudh_chawla@LUCIF3R:~$
```